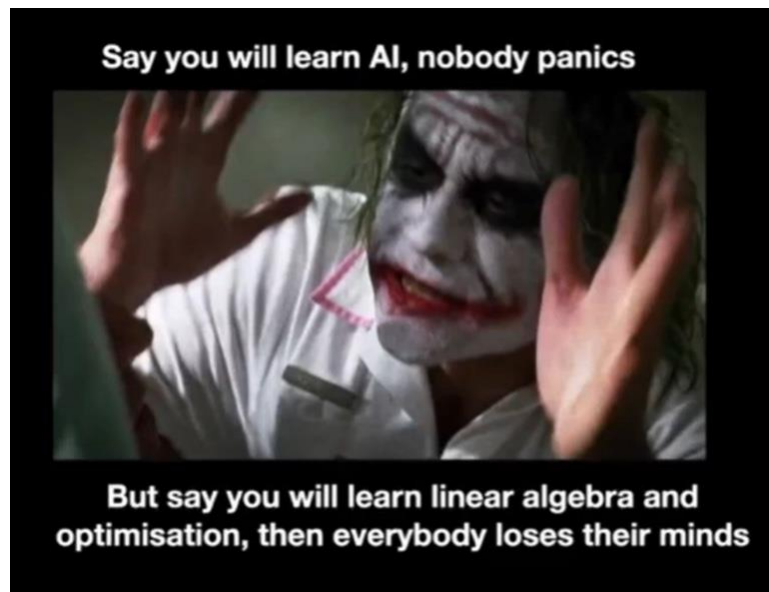


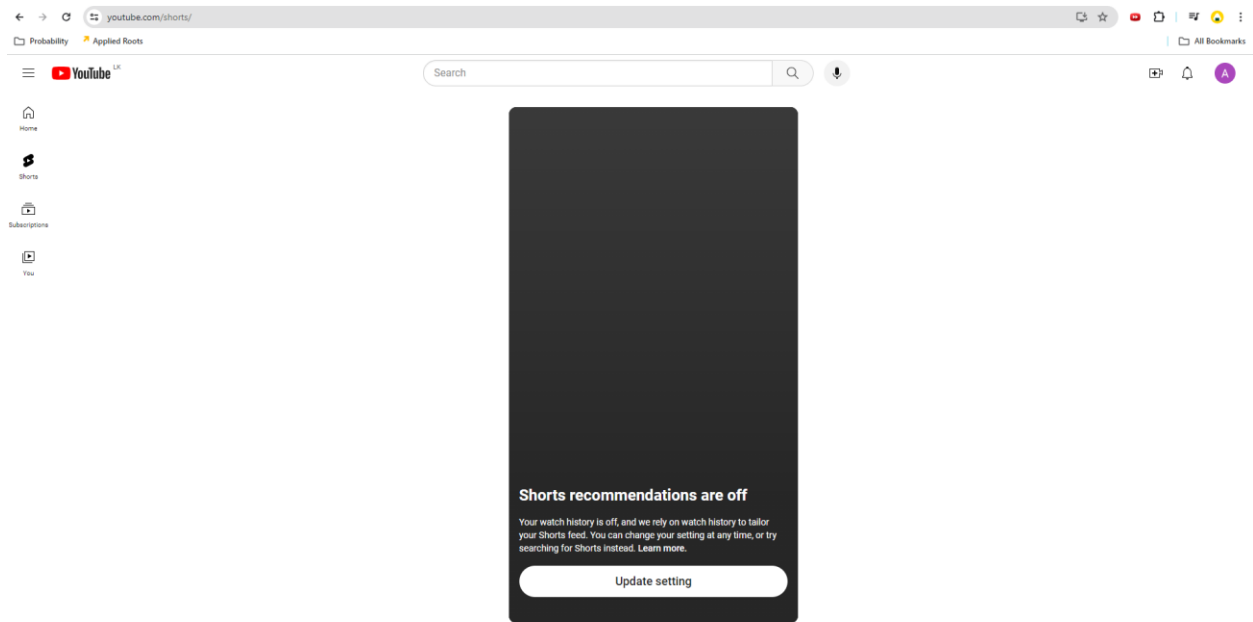
Linear Algebra 1 - The ML context



The flow would be like Concept -> Visualization -> Maths -> Code

Introduction to a real life problem





- Notice how each fish is getting sorted into different buckets.
- Inherently, we know that there must be a scanner that analyses certain parameters.

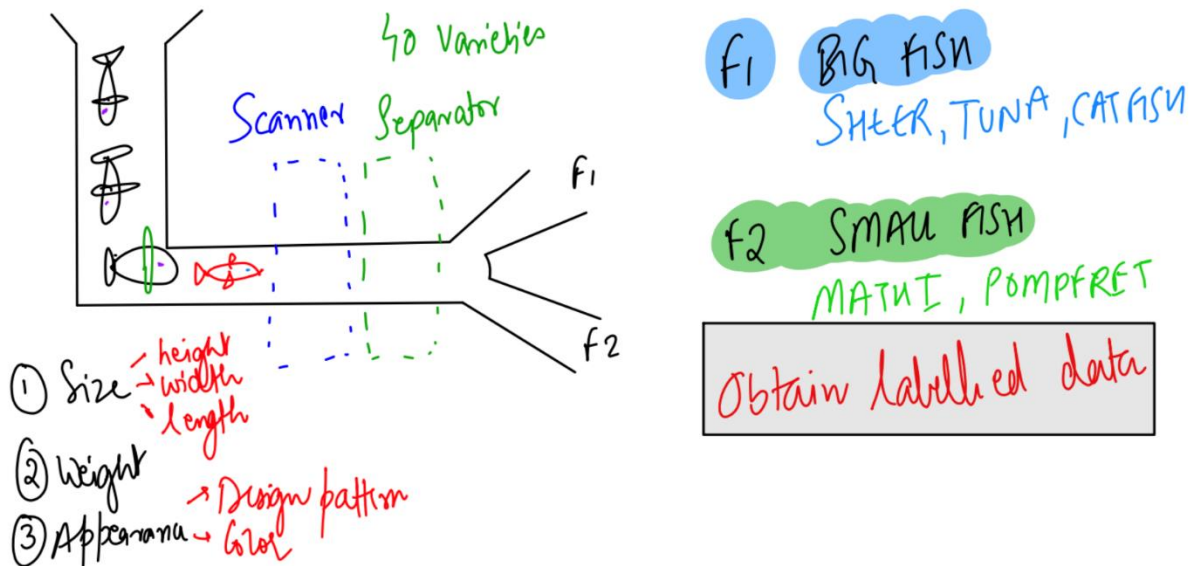
Q. What do you think are the parameters based on which the sorting works?

- Weight
- Size
- Appearance

- Colour
- Shape and so on

Machine Learning Applications

Example 1: Fish Sorting Machine



To solve this automation task we would need the scanner to **collect the data** send it somewhere (essentially a model), then the model make a prediction using the data and will return the output to the separator.

Terminologies

- So far what they did in previous modules was about analyzing the data given to them but here the task is to predict a label using the data.

Let say that we are given a **Labelled dataset** ie a table as shown below

Terminologies:

DATASET → features/attributes

Record/
datapoint

Width	Length	Weight	Type
30	50	80	1
11	23	28	2
29	43	72	1
16	31	36	2

TARGET
VARIABLE
LABELS dependant

Independent

Q. What do you call these parameters that help us determine the fish type?

- It is called as **features**.
- They are also referred to as **Independent variable**
- We can record values of each features using a sensor.

Q. And this different type of fishes is known as?

- These are known as **targets** or is also called as **labels** or **Dependent Variable**.
- It tells us which type does a given row belong to

Now given a datapoint our job is to predict it's respective Label

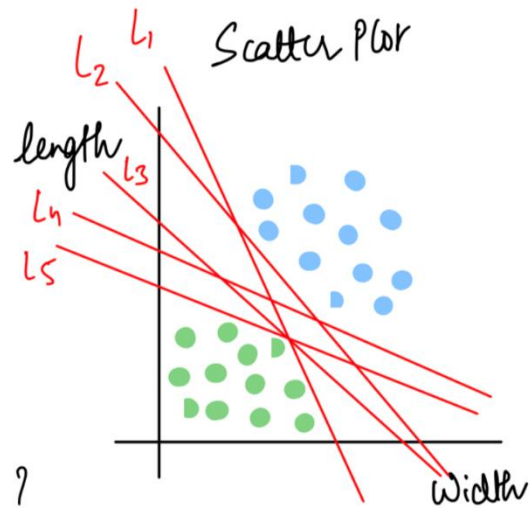
Visualizing the data

So let's take the data that we have and plot them

Visualisation:

Width	length	Weight	Type
30	50	80	1
11	23	28	2
27	43	72	1
16	31	36	2

∞ lines b/w Green & Blue point
Q. Which line is the Best one?



- Since we have 3 features with us let's pick up 2 features which is height and width and create a scatter plot to understand what is their relationship with each other as well as their combined relationship with the target variable.
- This is a 2-D plot
- As we can see that we can have infinite number lines between green points and the Blue points

Q. The question is which is the best line that we should pick that will help us separate Blue points from green points?

- This is the question that we will be solving in this entire module.

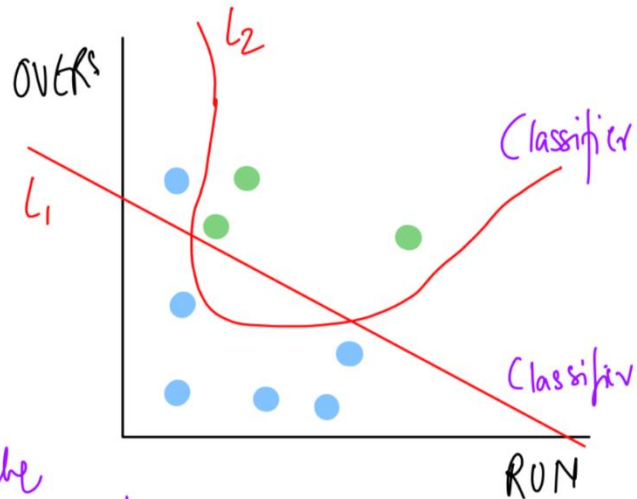
More ML Applications

IPL Win Prediction

Machine Learning Applications

Example 2: IPL Win Prediction.

RUNS	OVERS	OUTCOME
90	6	WIN
90	15	LOSS
18	1	WIN
30	7	LOSS



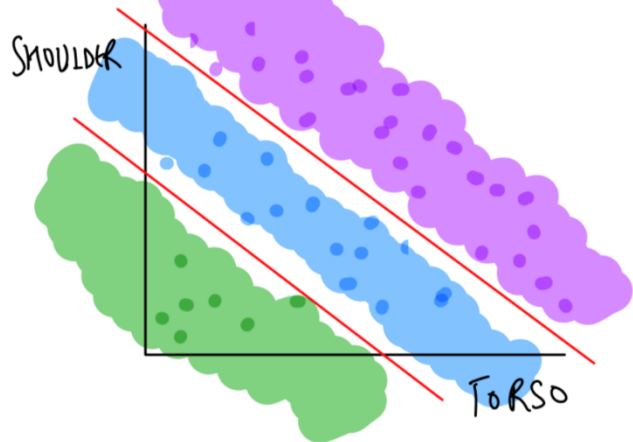
* Straight lines might not be the best option everytime.

T-Shirt size prediction

Machine Learning Applications

Example 3: T Shirt Size Predictions

TORSO	SHOULDER	SIZE
61	40	S
63	42	M
64	44	L
62	41	S
64	43	M
69	45	L



Multi Class Classification

Process of building an ML Algorithm.

a) Data Collection → labelled data

b) Data Visualisation 
PLR
tSNE, PCA

c) Choosing an appropriate geometrical structure to separate class

d) Choosing a LOSS function which helps decide the 'best' structure

e) Training / optimisation

Basics of Lines

The basic equation of Line is

$$y = mx + c$$

This is also referred to as **Slope intercept equation** of a Line

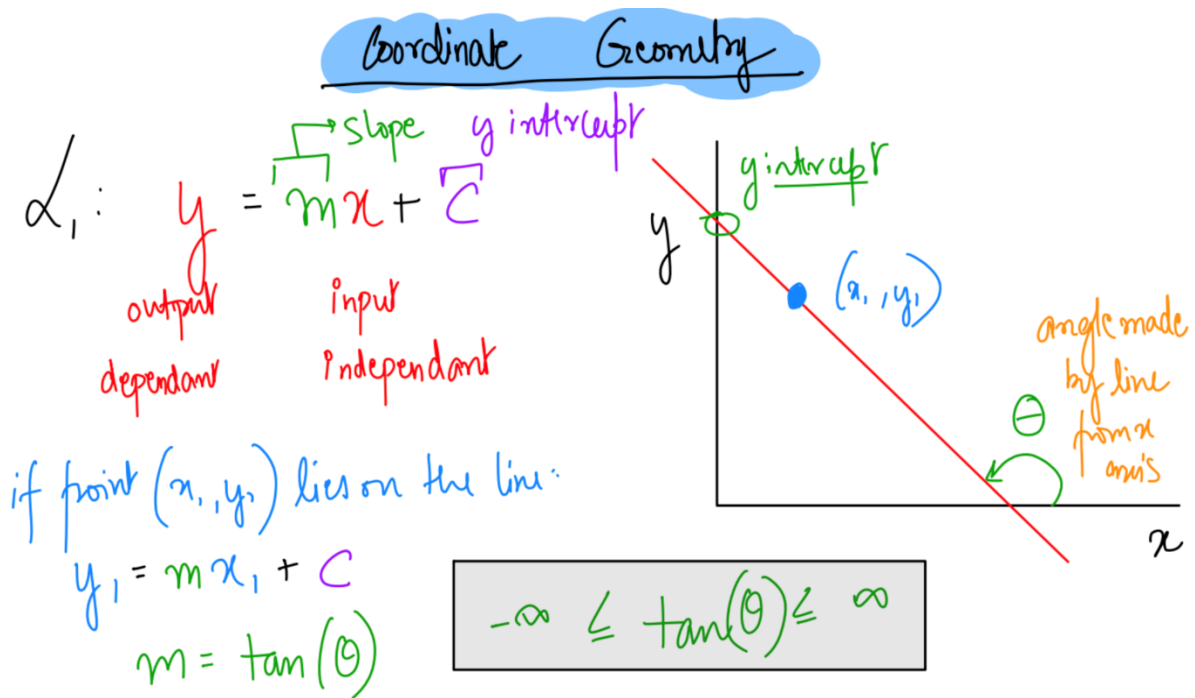
- **x** is input and is an independent variable
- **m** is slope of the line ie the angle it makes with the x axis
- **c** is the y-intercept
- **y** is the output and is a dependant variable ie it is dependent on x given that we keep m and c constant

For a given point (x_1, y_1) if it lies on our line L_1 it should satisfy the equation of our line and the equation would be

$$y_1 = mx_1 + c$$

The slope **m** can be calculated by calculating $\tan \theta = \frac{y_2 - y_1}{x_2 - x_1}$ where (x_1, y_1) (x_2, y_2) are two coordinates on the line.

- Range of $\tan \theta$ is from $-\infty$ to ∞
- $\tan(90) = \infty$ due which there is a limitation which is using the slope intercept form we cannot create a line which is 90 degrees to x-axis and to overcome this we use general form of line



General form of Line

We might have already studied this form which is

$$Ax + By + c = 0$$

- Now here instead of using A and B we are using W1 and W2
 - In the ML context they are referred to as **weights** that the computer learns
- Instead of x and y we use x1 and x2 they both mean the same thing that it has 2 Dimensions
 - In the ML context they are referred to as **features** which we saw above like length, weight, etc.
- C is represented as W0
 - In the ML context this is referred to as **Bias**
- We are essentially changing the way it is represented that is because we follow this form of representation in Machine learning and the terminologies change when we go to the Machine learning side of representation

We can convert general form of line into slope intercept form.

Line Equation: General form:

$$L_g : w_1 x + w_2 y + w_0 = 0.$$

In pink: Parameters: these are numbers which we expect the computer to learn automatically using optimization.

How to go from L_g to L_i :

$$w_1 x + w_2 y + w_0 = 0$$

$$w_1 x + w_2 y = -w_0$$

$$w_2 y = -w_0 - w_1 x$$

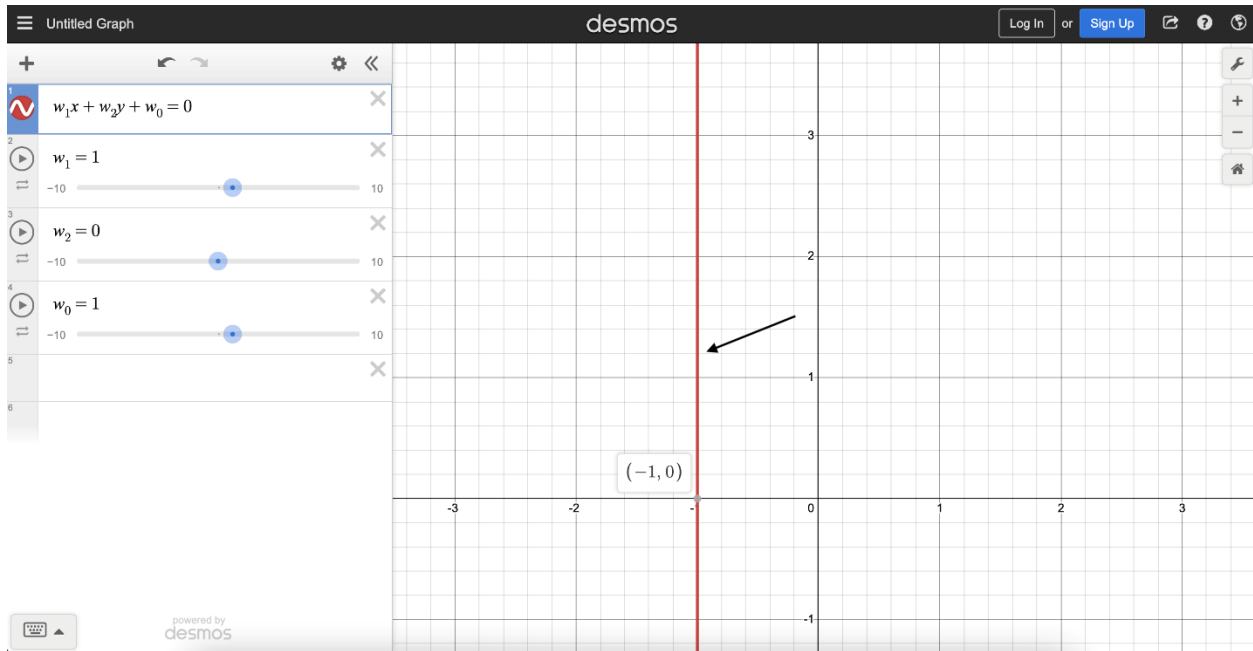
$$y = -\frac{w_0}{w_2} - \left(\frac{w_1}{w_2}\right)x$$

$$m = -\frac{w_1}{w_2}$$

$$c = -\frac{w_0}{w_2}$$

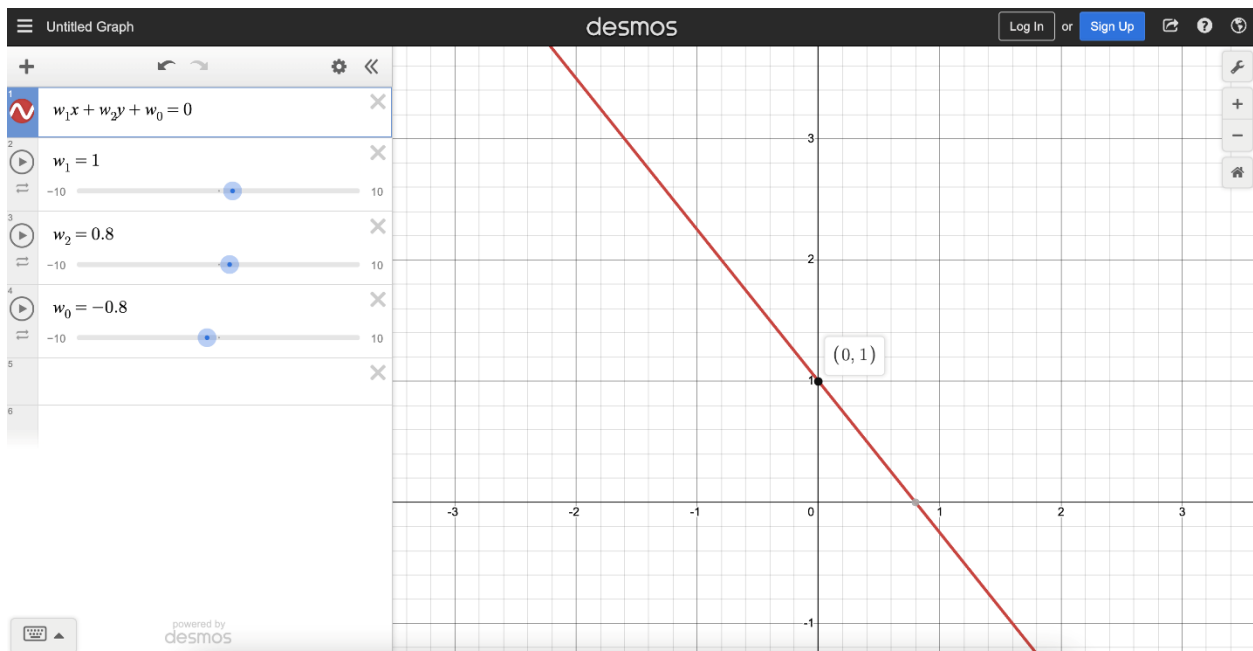
Now using the equation of m that we got after converting general form into slope intercept form

- if we want to have a perpendicular line to x axis we know that m has to be infinite and we can do that by equating w_2 to 0 and we can see that we have a line that is perpendicular to x axis



Now if we want the y intercept to be equal to 1 using the new equation of the intercept which is

$-w_0/w_2$ where if $w_2 = 0.8$ then w_0 should be equal to -0.8 .



Parallel Lines

If 2 lines are parallel to each other then they will have same angle with the x-axis which means that both the lines will have same slopes

Parallel lines

$$\theta_1 = \theta_2$$

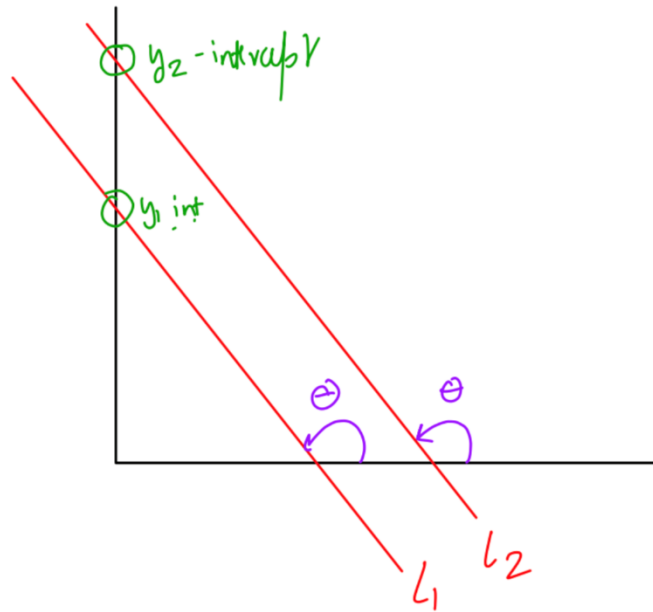
$$\tan(\theta_1) = \tan(\theta_2)$$

$$m_1 = m_2$$

$$w_1x + w_2y + w_0 = 0$$

$$w_3x + w_4y + w_5 = 0$$

$$-\frac{w_1}{w_2} = -\frac{w_3}{w_4}$$



Perpendicular lines

- The product of the slopes of 2 perpendicular lines is equal to -1
- if slope of line1 is m_1 and slope of line2 is m_2 then the formula to represent slope of 2 perpendicular lines is $m_1 * m_2 = -1$ which is equal to $m_1 = -1/m_2$ and $m_2 = -1/m_1$
- using the general form we have

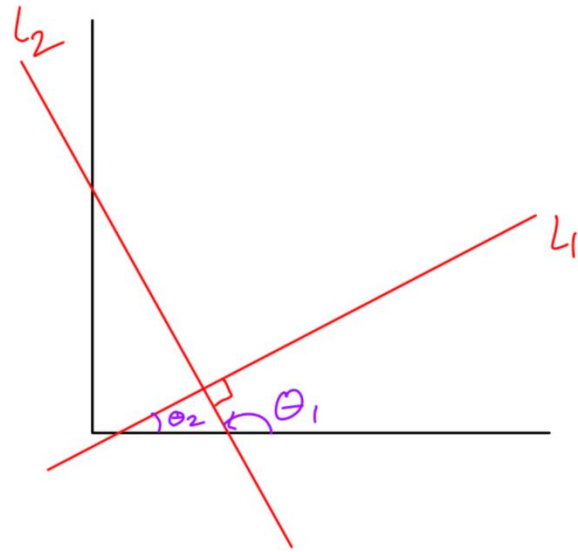
○ $-w_1w_2 * -w_3w_4 = -1$

Perpendicular lines

$$m_1 * m_2 = -1$$

$$\frac{+w_1}{w_2} x + \frac{w_3}{w_4} = -1$$

$$\begin{matrix} 2 & -1 \\ 1 & 2 \end{matrix}$$



More than 2 dimensions

So this is called as the **Generalized equation of a line**

- 2D it is a Line
- 3D it is a plane
- 4D it is 4D hyperplane
- nD it is a nD hyperplane

The term hyperplanes because we cannot visualize beyond 3D

What will we do if we have > 2 dimensions?

2D $w_1x_1 + w_2x_2 + w_0 = 0$ 2D Hyperplane Hyperplane

3D $w_1x_1 + w_2x_2 + w_3x_3 + w_0 = 0$ 3D Hyperplane

4D $w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + w_0 = 0$ 4D Hyperplane

nD $w_1x_1 + w_2x_2 + w_3x_3 + \dots + w_nx_n + w_0 = 0$
nD Hyperplane.

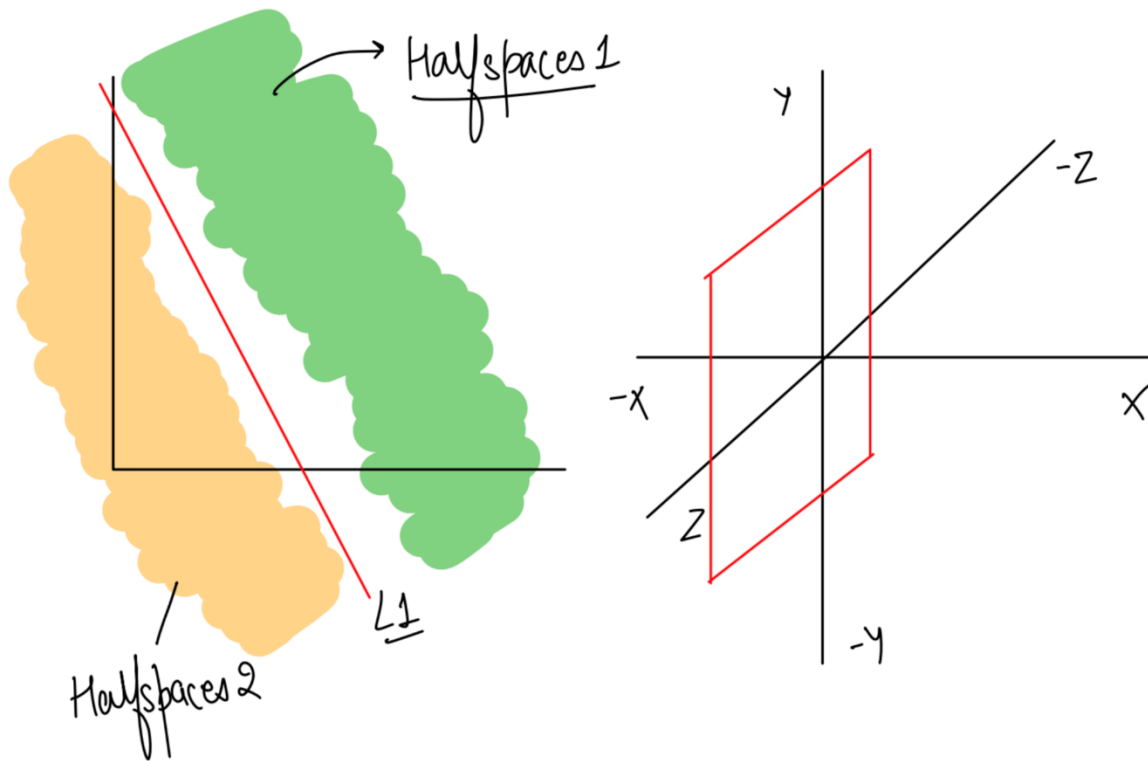
When we draw a line we can see that we are essentially dividing the 2D space into 2 different spaces

- one which lies above the line
- other is the one that lies below the line

Similarly in 3D we can divide the space into 2 parts using a Plane

- One which lies above the plane
- other is one that lies below the plane

And we can extend this concept till n dimensions



Why do we care about halfspaces?

- Whenever we are asked to classify a new datapoint we just plot the given datapoint and see in which half space does it belong to using which we can classify the given point.
- More on this we'll be clearing in the next class